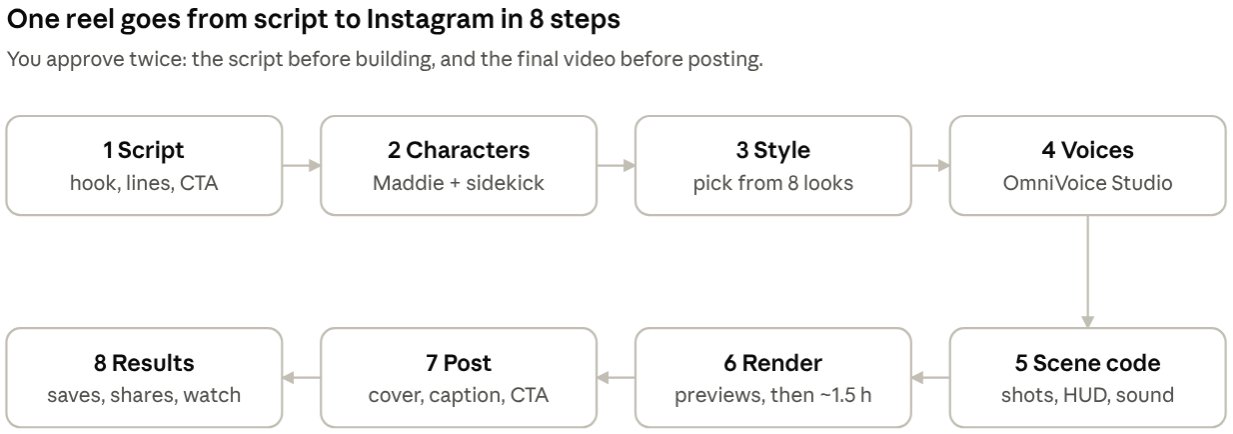
How to Make a 3D AI Reel in Code — Start to Finish

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Overview

Each reel is made entirely in code, with no camera and no editing app: an AI (Claude) writes the script and the 3D scenes, a free voice tool makes the voices, and you review and approve. A 36-second reel takes about 3 hours end to end, half of it the render.



reel pipeline · 8 steps

Steps 1–3 decide what the reel says and looks like, 4–6 produce the video, and 7–8 get it posted and measure it.

Step 1: Idea and script

Every reel starts as a written script, and nothing is built until I say "build".

1. **Pick the idea.** One relatable moment or one useful thing (a prompt, a GitHub repo, a trick). Reels that give something away for free get saved and shared most.
2. **Write the hook first.** The first 2 seconds must say what's coming. Example: "Hey Claude… how many animation styles can you do?" A hook that hides the point loses people.
3. **Write the lines.** Maddie asks or reacts; Claude answers short and dry. 12–18 lines, about 30–40 seconds.
4. **Mark the emotions.** Tags like [sigh], [laughter], [surprise-oh], [surprise-wa], [question-en] go on the lines that need them.
5. **Mark the caption keywords.** Words in `*stars*` turn yellow in the subtitles.

6. **End with a one-word CTA.** "Comment CLAUDE", "Comment PROMPT". Never a sentence for viewers to type.
7. **Review and approve.** I read the script, ask for changes, then say "build".

Step 2: Characters and look

The two characters and the room are built once in code and reused in every reel.

- **Maddie:** a 3D cartoon boy made from shapes in three.js (a JavaScript 3D library). Rules: human-looking body with shoulders, arms and legs connected; small lip movement while talking; no blush dots.
- **The AI sidekick:** a round hover-bot with headphones and a glowing face. It blinks, winks, tilts and hovers.
- **The set:** a purple living room with a couch, popcorn bowl, curtains, fairy lights and a city window. Each reel can move to a new location when the story needs it.
- **Acting:** each line gets poses (look at Claude, chin on hand, lean in, smile) and camera shots (close-up, two-shot, push-in).
- **Screen rules:** full 9:16 frame with no black bars, captions high on screen with yellow keywords, no page handle or name tags on the video.

Step 3: Animation style selection

The same 3D scene can be redrawn in different styles, so a new look costs code, not a new set of models. The style is switched per shot by a shader (a small program that recolors every pixel after the 3D render).

#	Style	How it's made	Best for
0	Toy 3D (default)	Normal 3D render with soft lights and bloom glow	Every reel's base look
1	Pixel art	Screen cut into big pixels, limited colors, game HUD (score, lives)	Retro or gaming jokes
2	Voxel	Characters rebuilt from little cubes, like Minecraft	Gaming, building themes
3	Comic book	Thick ink outlines, halftone dots, POW text	Loud, funny moments
4	Paper cut-out	Flat paper colors, white cut edges, tape strips	Soft, crafty, explainers

#	Style	How it's made	Best for
5	Neon	Black room, glowing outlines, grid floor	Night, tech, "wow" moments
6	Anime	Cel shading, bright hair, sparkles, speed lines	Emotional or hype beats
7	Tiny diorama	Whole room shrunk on a table, tilt-shift blur	Surprise or scale jokes
—	Code view	Green wireframe + the actual code typing on screen	"I made this with code" reveals

How I choose: pick the style that matches the line's emotion, keep one style per beat, and cut on the line where the style changes. A reel can stay in one style or switch on every line, like a "how many styles can you do?" reel.

Step 4: Voices

Voices come from OmniVoice Studio, a free, open-source voice app (an ElevenLabs alternative) that runs on your own computer with no API key. It can design a new voice from a description and clone it from a short clip. The cast used here: Maddie = American teen, the AI sidekick = young British.

1. **Install OmniVoice Studio** from its GitHub page and open it.
2. **Design each voice once.** In Voice Design, set gender, age, accent and pitch ("male, teenager, moderate pitch, American accent"), generate one sample line and save it.
3. **Clone from that saved sample every time.** Generate every line from the same saved clip in Voice Clone, so the characters sound identical in every reel.
4. **One file per script line.** Name them in order (01.wav, 02.wav…). Add emotion where the script marks it: a sigh, a laugh, surprise, a question tone.
5. **Clean up.** Trim the silence at both ends of each clip. Add a style effect where it fits: crunchy 8-bit sound for pixel art, echo for neon, a tiny high voice for the diorama.
6. **Listen and fix.** Regenerate any line that sounds rushed or flat, slower or reworded. 18 lines take about 10–15 minutes.

Step 5: Building the scene in code

Each reel is one scene file: a lines file (the script) plus a body file (everything you see and hear). Claude writes both.

1. **Timeline from the voices.** The real length of each voice clip sets when every line starts, so cuts and effects are tied to the words, not to guessed seconds.
2. **Shots.** Each beat gets a camera shot; a cut lands on the line that changes the style or the mood.
3. **Acting.** Poses and expressions are set per line: Maddie looks at Claude, leans in, smiles; Claude winks or tilts.
4. **Motion graphics (the HUD).** Titles, style badges (1–7), counters (7 → ∞), code on screen, the "PICK YOURS" wall and the comment card are drawn on top of the 3D.
5. **Captions.** Each line becomes a subtitle placed high on screen, keyword in yellow.
6. **Sound.** Pops, whooshes, coins and bass hits on the cuts, background music that dips under key lines and goes quiet for the big turn.

Step 6: Previews and the final render

The video is drawn frame by frame in a hidden Chrome browser, then joined with the voices, music and sound into one MP4.

1. **Still previews.** A few frames at the busiest moments are drawn and checked first: captions, badges, nothing overlapping.
2. **Line preview.** One frame per line, with voices, to check timing and that every graphic shows when it should (for example, ∞ appears before the code reveal).
3. **Full render.** Every frame at 1080 × 1920, 30 fps. A 36-second reel = about 1,090 frames at about 5 seconds each, so **about 1.5 hours**.
4. **Check the result.** A contact sheet (one frame every 2.3 seconds) confirms every section before you watch the MP4.

Important: changing only the voices still needs a full render, because the new clip lengths shift every cut, caption and mouth movement.

Step 7: Posting

Post the MP4 straight from the Instagram app as a Reel.

1. **Watch it once on your phone.** Check sound levels and that the captions are readable at phone size.
2. **Cover.** Pick the frame with the hook title so the grid shows what the reel is about.
3. **Caption.** First line repeats the hook, then a short list of what's inside, then the one-word CTA ("Comment CLAUDE and I'll send you all of them").
4. **Hashtags.** 3–5 specific ones (#animationstyles, #3danimation, #claudeai), not 30 generic ones.

5. **Time.** Post when your audience is online; midday and 7–9 PM work well in India.
6. **Deliver the freebie.** Send what you promised to everyone who comments the word.

Step 8: Checking results

Open the reel's Insights after 24 hours and compare it with your usual numbers. These four matter most:

Metric	What it tells you
Saves	People want to come back to it: the strongest push signal
Shares	People sent it to friends: reach beyond your followers
Avg watch time	Whether the hook and pacing held them (aim for over half the reel)
Comments	Whether the CTA worked

One example: a reel with a free prompt giveaway got 13,000 views, 554 saves and 201 shares, against about 1,500 views on a normal reel, with average watch time more than doubled.

Time and tools for one reel

Stage	Time	Tool (all free)
Script	15–30 min with review	Claude
Voices	10–15 min	OmniVoice Studio
Scene code + previews	30–60 min	Claude, three.js
Full render	about 1.5 h	Chrome (headless) + ffmpeg
Posting	10 min	Instagram app

10 script ideas to start with

Each idea follows the same pattern: a question or problem in the hook, a quick visual payoff, and a free giveaway behind a one-word comment.

#	Idea	Hook line	Payoff	Best style	CTA word
1	One prompt, five looks	"Can you make me look like a movie poster?"	The character flips through 5 poster styles, one per beat	Comic, neon, anime	POSTER
2	Resume in 60 seconds	"300 applications. Zero calls. Why?"	The AI rewrites one bullet on screen, before vs after	Toy 3D + code view	RESUME
3	Build a website by talking	"Can you build my portfolio while I eat?"	A site builds itself on screen, bite by bite	Toy 3D, then code view	SITE
4	Faceless channel in a day	"This channel has no face and 1 million views."	The AI shows the 4 tools behind it, one per cut	Pixel art (gamer feel)	FACELESS
5	Study plan for exams	"Exam in 7 days. I know nothing."	A 7-day plan stacks up as blocks, day by day	Voxel	PLAN
6	Excel to dashboard	"My boss wants a dashboard by 5 PM."	A messy sheet turns into a clean chart	Paper cut-out	DASHBOARD
7	Shrink a long video into Shorts	"I have a 1-hour podcast. Make it 10 reels."	The long bar splits into 10 small clips	Tiny diorama	CLIPS
8	Talk to your PDF	"300-page textbook. One question."	The book opens and the answer glows on the page	Neon	PDF
9	Cold message that gets replies	"Why does nobody reply to my DMs?"	Bad message vs 3-line good message, side by side	Comic (red X, green tick)	REPLY
10	"Is this real or AI?" game	"Guess which one is AI."	3 rounds, the reveal is the last line: all of it was code	Switches every round	GUESS

Tips that apply to all ten: keep it under 40 seconds, show the result before explaining it, and make the giveaway something people would save (a prompt, a checklist, a tool list).